# Keeping Comparison Open: Interface Scaffolding Against Premature Accountability Convergence in Regulated Investigations

*Sora Kim*

Pharmaceutical Manufacturing Industry, South Korea

sorakim.lab@gmail.com

**Abstract** Root cause analysis (RCA) in pharmaceutical manufacturing environments is expected to produce causal explanations that are both accurate and accountable — yet investigations unfold under closure pressure that structurally favors premature convergence on a single explanation before alternatives have been meaningfully compared. This paper reframes this failure as an HCI problem: not of generating explanations, but of maintaining comparison under conditions that prematurely collapse it. I introduce **Premature Accountability Convergence (PAC)** as a structural failure mode that locates the problem not in individual cognition but in the interaction between closure pressure, documentation requirements, and institutional incentives, and I propose *comparison preservation* as a design principle and *AI as comparison enforcer* as an inversion of the dominant model of AI in investigation support. A two-study empirical program with $N = 20$ GMP-experienced participants establishes the empirical basis and demonstrates the design response: Study 1 (timed reflection protocol at three timepoints) shows that 15 of 20 participants produced single-cause, person-centered explanations initially — with confidence averaging 1 out of 5 regardless of experience level, indicating institutional pressure rather than epistemic certainty — but 18 of 20 broadened their accounts within ten minutes; Study 2 (AI-supported RCA Reasoning Scaffolder with five structured stages) produced broader causal accounts in 18 of 20 participants, shifted 12 of 15 from person-centered to systemic explanations, and achieved usability ratings of 5.0 out of 5 and unanimous adoption intent, while all 20 participants simultaneously identified the audit trail implications of transparent investigation reasoning as a structural barrier to deployment.

**Keywords:** premature accountability convergence, root cause analysis, comparison preservation, AI scaffolding, GMP, closure pressure, regulated environments, human-AI interaction

# Introduction

Root cause analysis (RCA) in pharmaceutical manufacturing environments is expected to produce causal explanations that are both accurate and accountable. An investigation into a deviation should, in principle, compare multiple plausible explanations before arriving at a documented conclusion. In practice, investigations unfold under closure pressure — deadlines, documentation requirements, and procedural expectations that define when an investigation

must be considered complete. Under these conditions, reasoning does not fail to generate alternatives; it fails to maintain them in a comparable state long enough for systematic evaluation. Convergence on a single explanation occurs not because other explanations have been eliminated, but because comparison between them has collapsed.

This paper reframes this dynamic as an interaction problem. The issue is not that investigators fail to generate alternatives, but that the workflow does not sustain those alternatives in a comparable state long enough for meaningful evaluation. Interface design determines which explanations remain visible and which are suppressed before comparison is complete — establishing the conditions of investigation reasoning as a domain in which HCI intervention is both possible and necessary. The design question is not how to help investigators think better as individuals, but how to design workflows whose structure keeps comparison open against the institutional pressures that close it.

I conceptualize this dynamic as **Premature Accountability Convergence (PAC)**: the early stabilization of causal accounts that occurs when the comparison between alternative explanations collapses before reasoning has fully developed. PAC is not treated here as a cognitive bias or individual reasoning failure. It is treated as a structural outcome — a property of how investigative workflows organize, interrupt, and foreclose comparison under institutional pressure. This framing distinguishes PAC from prior accounts of premature closure in clinical diagnosis (**?**) and organizational decision making (Vaughan, 1996): those accounts locate the problem in individual cognition or organizational culture, and propose cognitive debiasing or cultural change as responses. PAC locates the problem in the interaction conditions that workflows create, and proposes redesign of those conditions as a tractable alternative. The distinction matters for design: if the problem is cognitive, interventions must change how people think; if the problem is structural, interventions can change the conditions under which thinking occurs.

I report a two-study empirical program in pharmaceutical quality systems. Study 1 establishes an empirical baseline using a timed reflection protocol with $N = 20$ GMP-experienced participants. Study 2 introduces an AI-supported RCA Reasoning Scaffolder that inverts the dominant model of AI in investigation support: rather than generating candidate explanations, the AI component detects convergence signals and requires their reopening — functioning as a comparison enforcer rather than an explanation source.

The contributions of this work are threefold. First, I introduce *comparison preservation* as a design principle for AI-supported reasoning systems in regulated environments. Second, I introduce *AI as comparison enforcer* as a design principle that inverts the dominant model of AI in investigation support. Third, I demonstrate empirically that comparison collapse is time-sensitive and addressable through design, and that structured scaffolding can delay convergence and broaden causal accounts.

# Related Work

## Causal Explanation as Organizational Outcome

The foundational premise of this study — that causal explanations in regulated investigations are shaped by the organizational conditions under which they are produced — draws on Vaughan's (1996) account of the Challenger launch decision. Vaughan shows that the engineers and managers at NASA reached a catastrophic decision through a series of incremental, locally reasonable judgments structured by production pressure, communication patterns, and the normalization of prior deviations. No single actor made an irrational decision; the catastrophic outcome emerged from the structure of how decisions were organized, not from individual failure. Applied to GMP deviation investigations, this framework implies that the causal explanations produced by RCA reflect the conditions of deliberation, not only the evidence available. An investigator who reaches a person-centered conclusion under closure pressure is not reasoning poorly; they are adapting rationally to conditions that the investigation workflow created. PAC names the specific mechanism through which this shaping occurs, and establishes it as a property of workflow structure rather than investigator reasoning.

## Premature Closure and Its Structural Form

Prior work on premature closure has focused primarily on cognitive mechanisms. **?** identifies premature closure as the most common cognitive error in clinical decision making, characterizing it as the failure to consider alternative diagnoses once an initial hypothesis has been accepted. The cognitive account is important: it establishes that convergence is a real and consequential phenomenon. But the cognitive account implies a cognitive response — debiasing training, structured checklists, metacognitive prompting. These responses treat the problem as one of individual reasoning quality and leave the structural conditions that produce convergence unchanged.

PAC offers a structural complement to this cognitive account. The mechanism is not that investigators fail to recognize the value of comparison; it is that the workflow does not sustain comparison as an interaction condition. In Study 1, confidence ratings averaging 1 out of 5 at the initial convergence timepoint establish that early convergence is not an expression of belief — investigators know they are converging prematurely. The barrier is structural: the documentation template does not require explicit comparison, the closure deadline creates pressure to stabilize, and the organizational incentive structure rewards documentation efficiency over investigative completeness. Addressing this structural form requires structural intervention, not cognitive debiasing.

## Recognition-Primed Decision Making and the Satisficing Problem

Klein's (1998) account of recognition-primed decision making (RPD) provides the cognitive background against which PAC becomes legible as a design problem. Klein demonstrates that expert decision makers under time pressure do not compare options — they recognize a situation, generate a single candidate course of action, and evaluate it before acting. This is adaptive in many high-stakes domains: RPD allows fast, effective action when comparison would be too costly. In regulated investigation contexts, however, expertise accelerates the very dynamic that investigation quality requires resisting: the more experienced the investigator, the faster the initial convergence, and the more confident the initial attribution — even when that confidence is not epistemically warranted. Study 1 confirms this pattern: confidence did not vary by experience level, but experienced practitioners converged more rapidly toward person-centered accounts at the initial timepoint. The design implication is counterintuitive: scaffolding that slows convergence may be more valuable for experienced investigators than for novices, precisely because their expertise makes premature closure more likely.

## Sensemaking and the Structure of Investigative Reasoning

Klein et al. (2006) extend the RPD account through a sensemaking framework that treats diagnosis as the progressive construction of a plausible account from incomplete evidence. Sensemaking is inherently comparative — alternatives are not just generated but maintained in relation to each other, evaluated against shared evidence, and progressively narrowed as the account becomes more robust. This comparative structure is what investigation quality requires. Yet the conditions of regulated investigation systematically suppress it: the documentation template demands a single root cause; the closure deadline compresses the time available for comparison; the review process evaluates the adequacy of the documented conclusion, not the adequacy of the comparison that produced it. PAC is, in these terms, a premature termination of sensemaking — not a failure of sensemaking capacity, but a structural truncation of the process before it can produce an account that reflects genuine comparison.

## Situated Action and the Conditions of Reasoning

Suchman's (1987) account of situated action establishes that reasoning is shaped by the immediate conditions of its execution. The investigation workflow is not a neutral container for reasoning; it is an active shaper of which explanations become salient, which alternatives remain visible, and which accounts are available for documentation. An investigation workflow that requires a documented root cause before alternatives have been meaningfully compared creates conditions that structurally favor premature convergence, independent of individual

reasoning quality. The scaffold in Study 2 is not a cognitive intervention — it is a redesign of the situated conditions of reasoning. By requiring explicit hypothesis generation before narrowing, surfacing convergence signals at the stage level, and posing verification questions before closure is permitted, the scaffold restructures the interaction conditions under which reasoning unfolds.

## Invisible Work and the Accountability Stakes of Comparison

Star and Strauss (1999) show that work performed in the gaps of formal systems tends to become invisible. In regulated investigation workflows, the comparative work of holding multiple causal explanations in parallel is precisely this kind of invisible work: not required by the documentation template, not rewarded by closure expectations, and not visible in the final investigation record. When an investigator considers and then sets aside a systemic explanation in favor of a person-centered one, the consideration leaves no trace. The final record reflects only the conclusion, not the comparison that did or did not produce it. This invisibility has accountability consequences: the audit trail documents the explanation that was adopted, not the alternatives that were foreclosed. The scaffold's pre-closure verification step addresses this invisibility directly, requiring documentation of which alternatives were considered before the final attribution is permitted.

## Human-AI Interaction and Comparison-Preserving Design

The AI component of the Study 2 scaffold reflects Amershi et al. (2019)'s principle that AI systems should support human control and avoid pre-empting human judgment. In regulated investigation contexts, the stakes of this principle are particularly high: any system that generates causal conclusions risks displacing the human judgment that regulatory accountability requires, and any system that narrows the comparison space risks amplifying the very dynamic that PAC names. The AI component is therefore constrained to a specific function: detecting signals of premature convergence and triggering prompts that reopen the comparison space. It does not suggest causes, evaluate hypotheses, or generate conclusions. This constraint is not merely a safety preference; it is a design commitment derived from the analysis of PAC as a structural problem. If the problem is that workflows close comparison prematurely, the appropriate AI role is to detect closure and require its reopening — not to generate a better set of alternatives for the investigator to converge on.

Buçinca et al. (2021) demonstrate that cognitive forcing functions — design interventions that interrupt automatic processing and require deliberate evaluation — reduce over-reliance without sacrificing efficiency. The PAC detection mechanism in Stage 3 of the scaffold extends this principle to a context where the risk is not over-reliance on AI outputs but over-reliance on the investigator's own first account. The forcing function does not slow reasoning arbitrarily; it

intervenes specifically when convergence-oriented language signals that comparison has begun to collapse, and requires the investigator to surface and evaluate alternatives before proceeding.

### AI-Assisted Decision Making and Interface Effects

Wang and Yin (2021) demonstrate that explanation interfaces alter how decision makers engage with AI outputs, with effects contingent on domain expertise — findings that establish investigation interfaces as active shapers of reasoning rather than neutral displays. The design implication is that the structure of the interface, not only its content, determines whether AI support expands or constrains the comparison space. Green and Chen (2019) show that algorithm-in-the-loop settings systematically fail to produce reliable decision making even when users are instructed to exercise independent judgment, establishing that structural conditions of human-algorithm interaction determine whether human interpretive authority is genuinely preserved. These findings ground the design logic of comparison preservation: if interface structure shapes what explanations become salient, redesigning that structure is a tractable intervention against premature convergence — and one that does not require changes to how investigators reason, only to the conditions under which their reasoning unfolds.

## Method

### Study Context

This study was conducted in the context of pharmaceutical manufacturing quality systems operating under GMP regulations. Deviation investigations in these environments are formal quality events requiring documented root cause analysis, CAPA planning, and qualified reviewer approval before closure. The closure pressure that shapes investigation reasoning is not metaphorical: it is encoded in regulatory expectations, procedural timelines, and quality system metrics that measure time-to-closure as an indicator of quality system performance. This structural pressure operates on investigators whether or not they are consciously aware of it, and shapes how reasoning develops before alternatives have been fully compared.

The pharmaceutical GMP context was selected not as a convenience sample but as a domain in which the conditions the framework identifies — closure pressure, documentation requirements, accountability consequences, and the validated-system constraint — are particularly pronounced and formally codified. The findings are therefore not generalizations from a convenient population; they are analyses of a domain in which PAC operates with unusual clarity and consequence.

## Participants

Both studies were conducted with the same participant pool of $N = 20$, recruited from the same pharmaceutical quality team, with direct experience in GMP deviation investigation workflows including RCA, CAPA documentation, and quality review. Experience levels ranged from entry-level to senior practitioners with five or more years in GMP quality roles. Recruiting from a single expert team was a deliberate methodological choice: it ensures consistent baseline familiarity with GMP investigation conventions and eliminates between-participant variance in domain knowledge as a confound. This design may have inflated acceptance ratings relative to a more diverse sample — a limitation acknowledged below.

Potential carry-over effects between studies are addressed through three design choices: a two-week minimum interval between studies; the scaffold's structural requirement to generate hypotheses not previously committed to, which creates a different reasoning task than the unstructured reflection of Study 1; and the fact that the primary outcome — a shift from person-centered to systemic attribution — is directionally opposite to what carry-over from Study 1 alone would predict. If Study 1 established a tendency toward person-centered attribution, carry-over would predict the same tendency in Study 2; the observed shift is therefore unlikely to reflect carry-over.

## Investigation Scenario

Both studies used the same standardized deviation scenario describing an out-of-specification result in an analytical testing context involving multiple instrument dependencies and procedural steps. The scenario was designed to support multiple plausible causal interpretations across person-centered, procedural, equipment-related, and organizational categories, such that no single explanation was obviously correct and all four categories could be legitimately supported by the scenario evidence. This design feature is essential for studying PAC: if the scenario had a clearly correct answer, early convergence would reflect good reasoning rather than structural pressure. The scenario was reviewed by two senior GMP practitioners for ecological validity before use.

## Study 1: Timed Reflection Protocol

Study 1 established an empirical baseline for how reasoning develops under closure pressure without structured support. The timed reflection protocol was chosen for three reasons. First, it captures the temporal dynamics of PAC directly: if comparison collapse is time-sensitive, a protocol that records reasoning at multiple timepoints can document the collapse as it occurs, rather than inferring it post-hoc from final documented conclusions. Second, it imposes realistic

closure pressure without simulating the full investigation workflow, allowing the dynamics of convergence to be observed without the confounds of documentation burden and reviewer interaction. Third, it produces directly comparable data across participants, since the timepoints and scenario are standardized.

Participants produced causal explanations at three timepoints (0, $+5$, $+10$ minutes) using a standard RCA template, recording causal type (person-centered, procedural, equipment-related, organizational, or mixed) and confidence rating (1–5 scale). Causal type classification was performed directly by participants via structured response options, reflecting professionally grounded judgment consistent with GMP documentation practice. This approach treats causal type as a practitioner judgment rather than an analyst inference, which is appropriate for a study whose central claim is that causal type reflects institutional conditions rather than evidence alone. Open-ended prompts at each timepoint invited participants to describe their reasoning and any changes in their account.

## Study 2: RCA Reasoning Scaffolder

Study 2 introduced an AI-supported RCA Reasoning Scaffolder implemented as a Streamlit web application. The scaffold was designed in response to the specific structural conditions that Study 1 identified as producing PAC: the absence of a requirement for explicit hypothesis generation before narrowing; the absence of a mechanism for detecting convergence-oriented language at the stage level; and the absence of a pre-closure check that requires documentation of which alternatives were considered. Each of the five stages addresses one or more of these conditions directly.

The design rationale for each stage follows from the PAC account. Stage 1 (Intake) captures deviation context without commitment to a causal conclusion, establishing that no hypothesis has been selected before the investigation has begun. Stage 2 (Hypotheses) requires generation of multiple causal candidates before any narrowing is permitted, and flags individual-blame language with a PAC risk rating — making convergence-oriented framing visible at the moment it occurs rather than after the fact. Stage 3 (PAC Analysis) applies rule-based convergence detection and activates an AI component when individual-blame language exceeds a threshold, surfacing alternative causal pathways and generating reopening questions that must be addressed before proceeding. Stage 4 (Reasoning) requires explicit per-hypothesis evaluation before any single explanation is selected, ensuring that the comparison stage is not collapsed by advancing to a conclusion before alternatives have been evaluated. Stage 5 (Pre-Closure) poses three verification questions before closure is permitted, requiring the investigator to document which alternatives were considered and confirm that the final account reflects reasoning rather than convenience.

1. **Intake** (Figure 1): structured capture of deviation context without commitment to a causal conclusion. A live hypothesis panel tracks active, narrowed, and dropped counts alongside a closure signal indicator.

2. **Hypotheses** (Figure 2): generation of multiple causal candidates across person, procedure, equipment, environment, and organizational categories. Each hypothesis receives a PAC risk rating (High/Low) based on convergence-oriented language detection. Investigators are instructed not to narrow yet.

3. **PAC Analysis** (Figure 3): rule-based detection of convergence-oriented language with prompts to surface underexplored contributors. An AI component activates when individual-blame language is detected, surfacing alternative causal pathways and generating reopening questions.

4. **Reasoning** (Figure 4): explicit per-hypothesis evaluation of contributing factors and evidence, with plausibility rating and investigation note for each hypothesis before any single explanation is selected.

5. **Pre-Closure** (Figure 5): a required check confirming alternatives were considered before final attribution, displaying hypothesis state and posing three verification questions before closure is permitted.

The AI component was integrated into Stage 3 only, functioning exclusively as a conditional trigger for reopening the comparison space. It did not suggest causes, evaluate hypotheses, or generate conclusions. This design constraint is not a limitation of the current implementation but a principled choice: an AI component that generates causal candidates would narrow the comparison space by making generated alternatives more salient than unaided ones, reproducing the very dynamic that the scaffold is designed to resist.

## Measures

Post-task assessment included five construct scales: usability (5 items), reasoning expansion (5 items), PAC mitigation (4 items), perceived usefulness (3 items), and blame pressure (1 item). All Likert items used a 5-point scale. Confidence change was assessed via pre/post 0–10 scale. Open-ended responses probed qualitative experience with the scaffold, with particular attention to participants' accounts of how the scaffold changed their reasoning process and their assessments of deployment feasibility.

# Findings

## Study 1: The Time-Sensitivity of Comparison

At the initial timepoint (0 minutes), 15 of 20 participants produced single-cause, person-centered explanations. Confidence ratings averaged 1 out of 5 regardless of experience level — a finding whose significance extends beyond the low number. The dissociation between convergence and confidence establishes that early attribution is not an expression of belief. Participants were not converging because they were certain; they were converging despite uncertainty. Post-task discussion confirmed this: all 20 participants described their initial explanation as selected for documentation acceptability rather than investigative completeness. Representative accounts included: "in a real investigation, you pick the explanation that closes the file. A systemic cause just opens more questions" and "I knew it probably wasn't just the analyst, but that's what goes in the template." These accounts establish that PAC is a rational adaptation to structural conditions, not a failure of reasoning quality.

With five additional minutes of unstructured reflection, 8 of 20 participants revised their accounts substantially, broadening causal type or explicitly expressing uncertainty about their initial attribution. By ten minutes, 18 of 20 had broadened their explanation or become more explicitly uncertain. This time-sensitivity is the central empirical finding of Study 1: comparison is not absent but suppressed, and modest temporal extension can partially restore it. The implication is that PAC is addressable through design interventions that extend or restructure the time available for comparison — not through changes to investigator reasoning capacity.

Two patterns in the timing data merit attention. First, the transition from initial convergence to broadened account was not continuous: most participants maintained their initial account through the five-minute mark and then revised substantially, suggesting a threshold effect rather than gradual development. Second, the participants who did not revise by ten minutes were disproportionately those with the most senior experience levels — consistent with the RPD account (Klein, 1998) and suggesting that expertise, paradoxically, amplifies structural convergence by making initial pattern recognition more fluent.

## Study 2: Scaffold Effects

18 of 20 participants produced broader causal accounts with the scaffold than they had produced in Study 1. 15 of 20 reconsidered their initial attribution after Stage 3; of these, 12 shifted from person-centered to systemic or procedural explanations. Post-task assessment confirmed the scaffold's effects across all measured constructs: usability averaged 5.0 out of 5; reasoning expansion averaged 4.8; perceived usefulness averaged 4.9; PAC mitigation averaged 4.7; blame pressure averaged 1.0 out of 5.

Confidence ratings were maintained or slightly decreased while causal breadth increased substantially — a dissociation that requires interpretation. In most reasoning contexts, expanded uncertainty would be a negative outcome. In high-accountability regulated contexts, it is precisely the outcome that makes adoption feasible: an investigator whose causal account has broadened while their confidence has remained stable has not become less capable of producing a defensible investigation conclusion. They have become more capable of producing one that reflects genuine comparison rather than structural pressure.

The usability ceiling is a domain effect rather than a measurement artifact. GMP investigation workflows have no existing analogue for structured comparison scaffolding; participants evaluated the tool against the absence of any structured support rather than against a competing interface. This context makes the ceiling informative: what participants rated at 5.0/5 was not a marginal improvement over an existing system but the first instantiation of a design category that did not previously exist in their domain.

All 20 participants expressed adoption intent. All 20 simultaneously identified the audit trail as a structural barrier to deployment. The co-occurrence of these two responses is the most consequential finding of Study 2, and it deserves careful attention. Participants were not ambivalent about the tool's value; they were clear that they wanted it. Their concern was organizational, not personal: "the problem isn't the tool. The problem is that the tool would show that we looked at the systemic cause and decided not to pursue it. That's a conversation no one wants to have." A second participant framed the barrier differently but identified the same mechanism: "right now, the investigation record shows what we concluded. This would show what we considered and set aside. That changes what the audit trail proves." These accounts establish that the barrier to deployment is not technical feasibility or user acceptance but organizational readiness to tolerate transparency about the choices that investigations currently make invisibly.

# Discussion

## Comparison Preservation as Interaction Design

Comparison preservation is a structural intervention, not a cognitive aid. It does not help investigators think better in the sense of improving their reasoning capacity; it changes the conditions under which their reasoning occurs. Prior HCI work establishes that interface structure actively shapes human judgment (Wang and Yin, 2021; Green and Chen, 2019) and that cognitive forcing functions reduce over-reliance on automatic processing (Buçinca et al., 2021). Comparison preservation extends this work into a more specific context: the risk is not over-reliance on AI outputs but over-reliance on the investigator's own first account, produced

under institutional pressure before comparison has been adequately sustained.

The distinction between cognitive aid and structural intervention matters for how the design is evaluated. If comparison preservation were a cognitive aid, its success would be measured by whether investigators reason better — whether their final accounts are more accurate, more complete, or more epistemically justified. If it is a structural intervention, its success is measured by whether the interaction conditions it creates are ones in which comparison can occur: whether alternatives are generated before narrowing, whether convergence signals are surfaced before they become irreversible, whether the comparison that did occur is documented before closure. Study 2 measures both, but the structural measures — causal breadth, shift from person-centered to systemic attribution, pre-closure verification — are the primary evidence for the design principle. The confidence and usability measures confirm that the structural intervention does not destabilize the practitioner; they are not independent evidence for the principle's validity.

## AI as Comparison Enforcer

The dominant model for AI in investigation support treats AI as a producer of candidate explanations. AI systems in this model analyze deviation data, generate likely root causes, and rank or recommend explanations for investigator review. This model inverts the problem that PAC names: it generates more alternatives for investigators to converge on rather than addressing the conditions under which convergence occurs. An AI that generates a ranked list of causal candidates does not preserve comparison; it provides a pre-structured comparison space that may make the investigator's own first account more salient if it appears at the top, or may introduce AI-generated accounts that are treated as authoritative rather than as candidates for evaluation.

The scaffold inverts this logic. The AI component's function is not to generate alternatives but to detect when comparison has collapsed and to require its reopening. This instantiates Amershi et al.'s (2019) judgment-preservation principle in a more demanding form than the guideline typically requires: in a high-accountability regulated context, preserving human judgment means designing AI to resist the institutional pressures that produce premature convergence, not merely to make its outputs transparent or its limitations clear. The AI component in Stage 3 does not tell the investigator what to conclude; it tells the investigator that the comparison conditions under which they are about to conclude have not been adequately sustained, and requires them to address that deficit before proceeding.

This design has a specific implication for how AI support in regulated environments should be evaluated. The standard evaluation framework for AI-assisted decision making asks whether AI support improves the accuracy or quality of final decisions. In comparison-preserving design, this is the wrong question: the goal is not to improve the final decision but to improve the conditions under which the decision is made. The appropriate evaluation asks whether the

interaction conditions the AI creates — the comparison space it sustains, the convergence signals it surfaces, the alternatives it requires to be evaluated — are ones in which genuine human judgment is possible. Whether that judgment produces better final decisions is an empirical question that requires longer-horizon study in live investigation settings.

## PAC as a Structural Problem

The finding that all 20 participants selected explanations oriented toward documentation acceptability rather than investigative completeness is evidence of rational adaptation, not poor reasoning. Participants were not making mistakes; they were responding correctly to the incentive structure they operate within. Confidence ratings averaging 1 out of 5 at the initial timepoint establish that convergence is produced by institutional structure, not by belief: investigators know they are converging prematurely, and they do so anyway because the alternative — a systemic investigation that opens rather than closes — is organizationally costly in ways that the documentation system makes invisible.

This extends Vaughan's (1996) account of the normalization of deviance: the explanation that gets documented is shaped by the accountability environment in which the investigator operates, not only by the evidence available. PAC is the mechanism through which accountability environments produce investigation conclusions: by structuring closure, organizing comparison, and rewarding documentation efficiency, they create conditions in which person-centered attribution is the rational output of a well-functioning quality system. The scaffold does not change the incentive structure; it changes the interaction conditions within which investigators respond to it, making comparison available as an option rather than leaving it dependent on individual resistance to institutional pressure.

## The Organizational Adoption Problem

All 20 participants expressed adoption intent; all 20 simultaneously identified the audit trail as a structural barrier. This combination is not contradictory; it reflects the gap between individual preference and organizational incentive. Individual practitioners want the conditions the scaffold creates — better investigation quality, reduced individual blame attribution, more defensible conclusions — because they bear the costs of the current absence of those conditions. Organizations may not want the transparency the scaffold produces, because that transparency makes visible choices that are currently made invisibly: the choice not to pursue a systemic explanation, the choice to prioritize closure over completeness, the choice to document a person-centered conclusion whose epistemic basis is weak.

The scaffold surfaces a condition that currently remains invisible: the choice not to pursue a systemic explanation is already being made in every investigation that closes with a person-

centered attribution. The scaffold does not create this choice; it creates a record of it. Whether regulated organizations can tolerate — or are required by their regulatory environment to produce — that level of transparency is an organizational and regulatory question that Regulatory HCI cannot answer through design. It requires engagement with quality culture, audit standards, and the regulatory frameworks that govern what investigation records must contain. The most important implication may be that Regulatory HCI tools are not supplements to existing compliance systems but challenges to the accountability assumptions embedded in those systems — and that deploying them requires organizational readiness to confront those assumptions, not merely technical capacity to implement the interfaces.

## Limitations and Future Work

The use of $N = 20$ participants from a single GMP quality team reflects a structural constraint of the domain rather than an arbitrary methodological choice. Competencies required for GMP investigation work are not transferable from adjacent populations: the domain knowledge, regulatory familiarity, and investigation experience that make participants' responses ecologically valid also limit the generalizability of findings to other quality teams and regulatory contexts. $N = 20$ represents the boundary of what domain-valid sampling permits within a single quality organization; larger samples would require multi-site recruitment, which introduces between-site variance in quality system design as a confound.

The synthetic investigation scenario is ecologically valid in structure but does not reproduce the time pressure, stakeholder dynamics, and documentation burden of a live investigation. Participants knew they were in a study, which may have reduced closure pressure relative to a real quality event. Future work should examine PAC in live investigation settings using observation and document analysis rather than controlled evaluation, to understand how the dynamics Study 1 identifies operate when investigators are under genuine regulatory pressure.

The scaffold has been evaluated in a single-session prototype study; its effects on investigation quality in sustained deployment are unknown. The organizational adoption barrier identified by all 20 participants is not merely a concern for future deployment — it is the most important finding of the study, and it requires organizational and regulatory research that extends beyond HCI methods. Future work should examine the conditions under which regulated organizations are willing to produce transparent investigation records, and whether regulatory frameworks can be designed to incentivize rather than penalize that transparency.

# Conclusion

This paper introduced Premature Accountability Convergence (PAC) as a structural failure mode in regulated investigation workflows and reframed it as an interaction problem concerning the conditions under which comparison is sustained or collapsed. PAC is distinguished from prior accounts of premature closure by its structural rather than cognitive framing: the problem is not that investigators reason poorly, but that the workflows in which they reason do not sustain comparison as an interaction condition against the institutional pressures that close it.

Study 1 established that comparison is time-sensitive and that early convergence reflects institutional pressure rather than epistemic certainty: confidence ratings averaged 1 out of 5 at the initial convergence timepoint, and all 20 participants described their initial attribution as selected for documentation acceptability rather than investigative completeness. Study 2 demonstrated that an AI-supported RCA Reasoning Scaffolder — with an AI component functioning as a convergence detector rather than an explanation source — produced broader causal accounts, shifted attribution from person-centered to systemic explanations, and achieved unanimous adoption intent, while surfacing a structural barrier to deployment that is organizational rather than technical.

Three contributions follow. First, comparison preservation as a design principle: the goal of AI-supported investigation systems is not to generate better explanations but to sustain the conditions under which comparison can occur. Second, AI as comparison enforcer as a design inversion: the appropriate AI role in high-accountability regulated contexts is to detect convergence and require its reopening, not to narrow the comparison space with generated alternatives. Third, the organizational adoption problem as a first-class research concern for Regulatory HCI: the gap between individual adoption intent and organizational deployment feasibility is not a technical limitation but a structural finding about the accountability assumptions embedded in existing compliance systems — and addressing it requires organizational and regulatory engagement that extends beyond interface design.

# References

Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Paul Collisson, Jina Suh, Shamsi Iqbal, Paul N. Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, and Eric Horvitz. Guidelines for human-AI interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, CHI '19. ACM, 2019.

Zana Buçinca, Maja Barbara Malaya, and Krzysztof Z. Gajos. To trust or to think: Cognitive forcing functions can reduce overreliance on AI in AI-assisted decision making. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW1):1–21, 2021.

Ben Green and Yiling Chen. The principles and limits of algorithm-in-the-loop decision making. *Proceedings of the ACM on Human-Computer Interaction*, 3(CSCW):Article 50, 2019.

Gary Klein, Brian Moon, and Robert R. Hoffman. Making sense of sensemaking 1: Alternative perspectives. *IEEE Intelligent Systems*, 21(4):70–73, 2006.

Gary A. Klein. *Sources of Power: How People Make Decisions*. MIT Press, 1998.

Susan Leigh Star and Anselm Strauss. Layers of silence, arenas of voice: The ecology of visible and invisible work. *Computer Supported Cooperative Work*, 8(1–2):9–30, 1999.

Lucy A. Suchman. *Plans and Situated Actions: The Problem of Human-Machine Communication*. Cambridge University Press, 1987.

Diane Vaughan. *The Challenger Launch Decision: Risky Technology, Culture, and Deviance at NASA*. University of Chicago Press, 1996.

Xinru Wang and Ming Yin. Are explanations helpful? A comparative study of the effects of explanations in AI-assisted decision-making. In *Proceedings of the 26th International Conference on Intelligent User Interfaces*, IUI '21, pages 318–328. ACM, 2021.

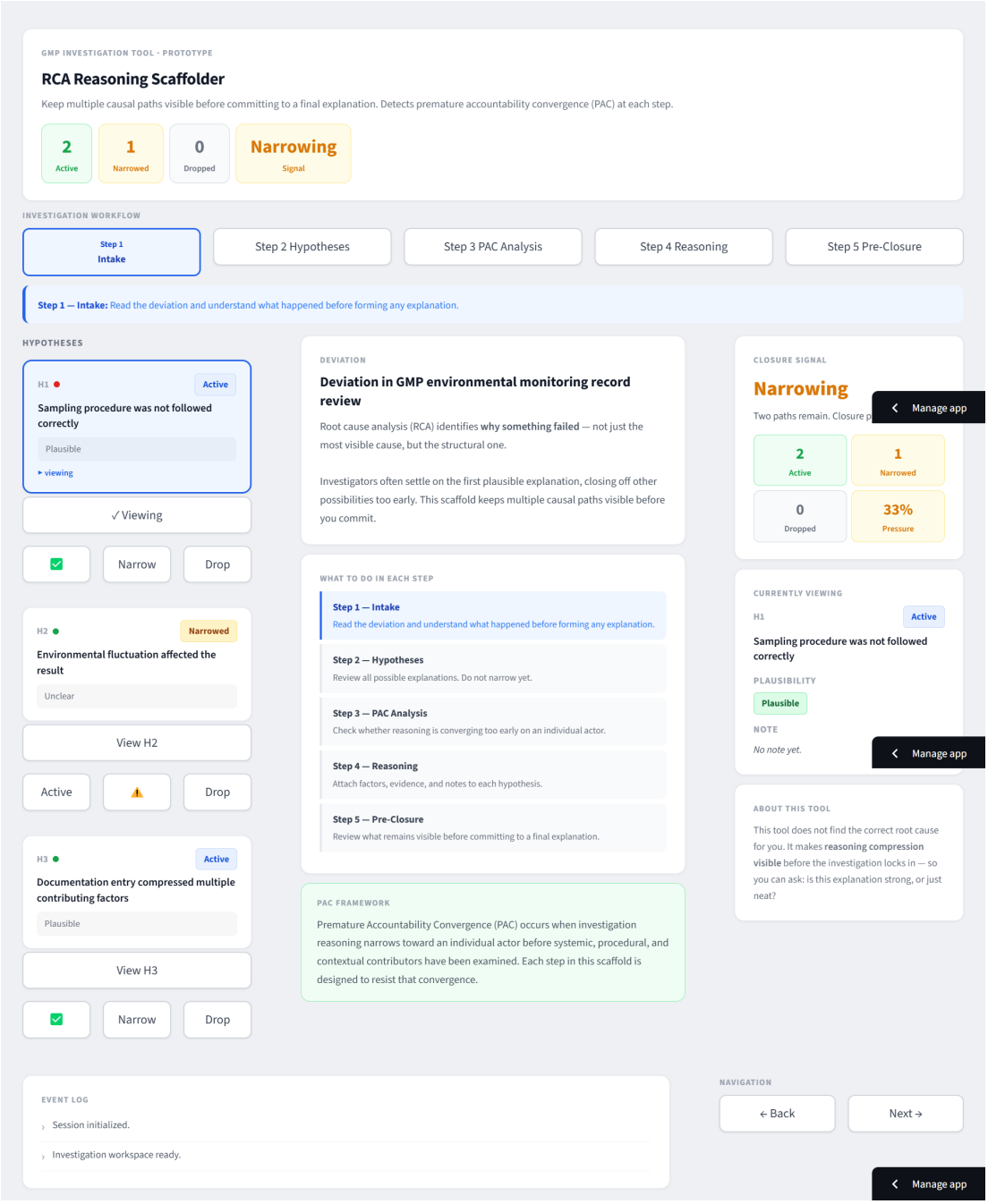



Figure 1: RCA Reasoning Scaffolder: Step 1 (Intake). The hypothesis panel shows three active hypotheses. The closure signal panel shows 2 active, 1 narrowed, 0 dropped, 33% pressure — indicating narrowing has begun before hypothesis expansion is complete.

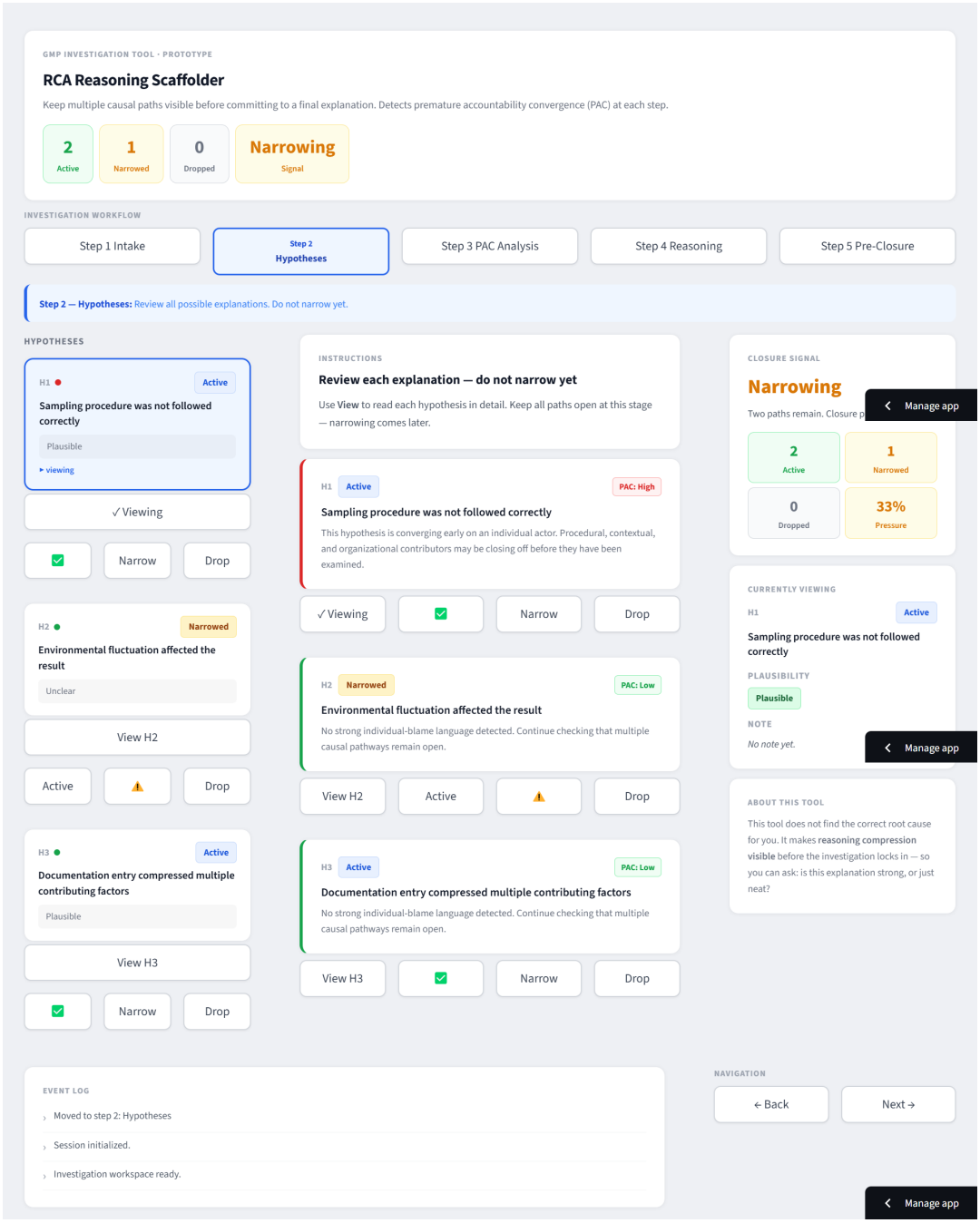



Figure 2: RCA Reasoning Scaffolder: Step 2 (Hypotheses). H1 is flagged PAC: High, indicating convergence-oriented language targeting an individual actor. The instruction "Review each explanation — do not narrow yet" enforces comparison preservation at the stage level.

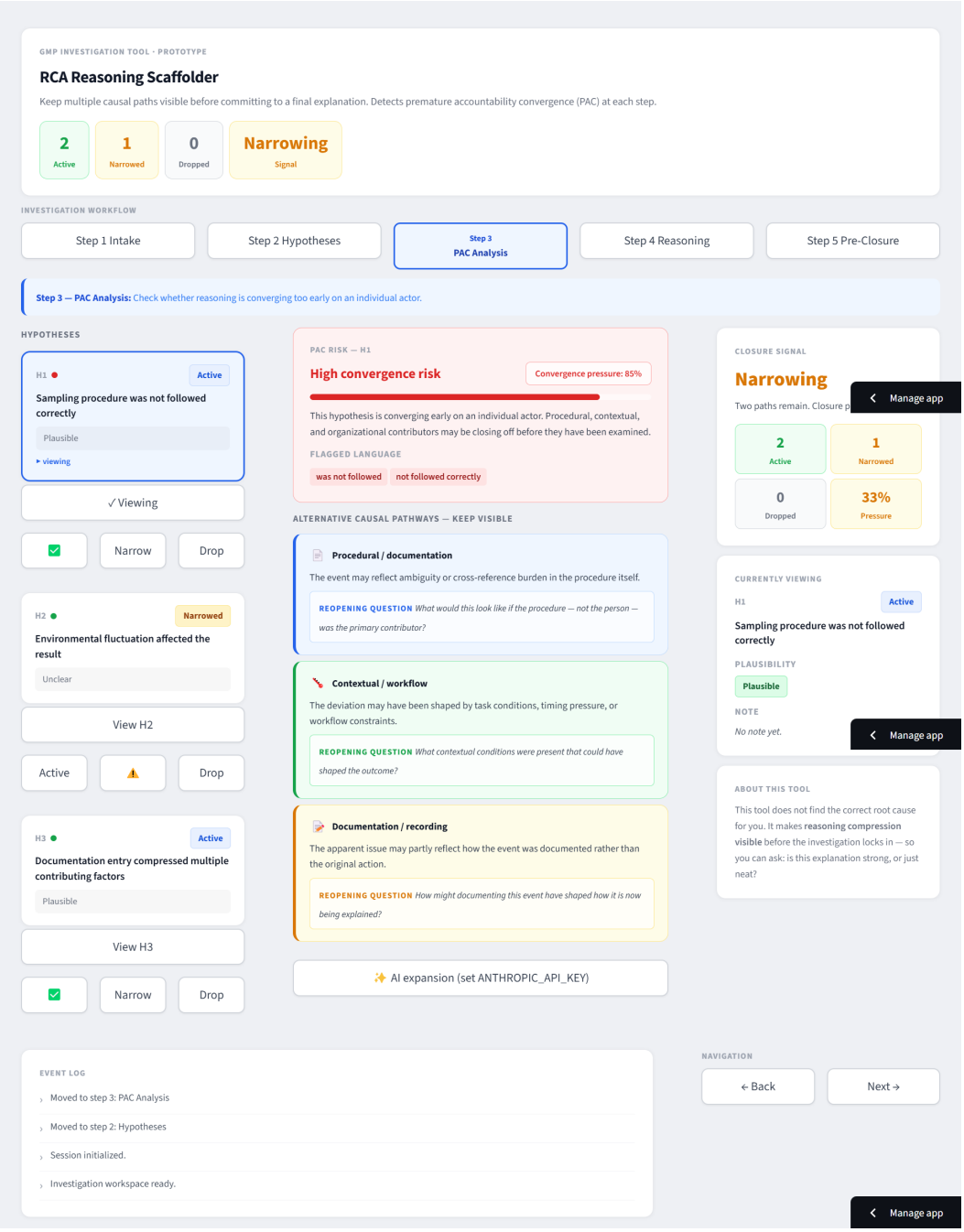



Figure 3: RCA Reasoning Scaffolder: Step 3 (PAC Analysis) with AI layer active. H1 is flagged as high convergence risk (85%) with individual-blame language tokens highlighted. The AI component surfaces three alternative causal pathways with reopening questions that must be considered before closure.

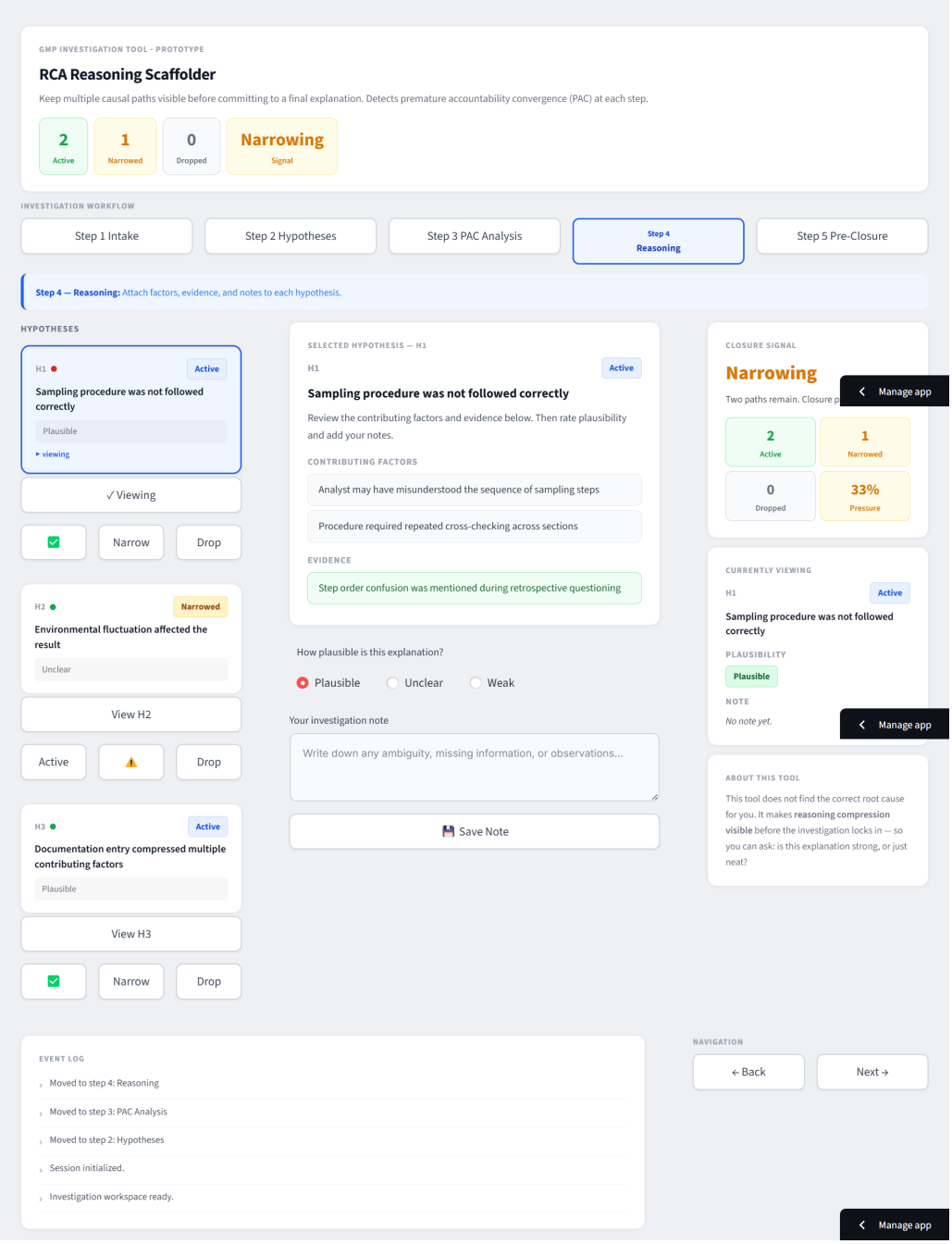



Figure 4: RCA Reasoning Scaffolder: Step 4 (Reasoning). Contributing factors and evidence for H1 are displayed for explicit evaluation. The investigator rates plausibility and adds investigation notes before moving to the next hypothesis.

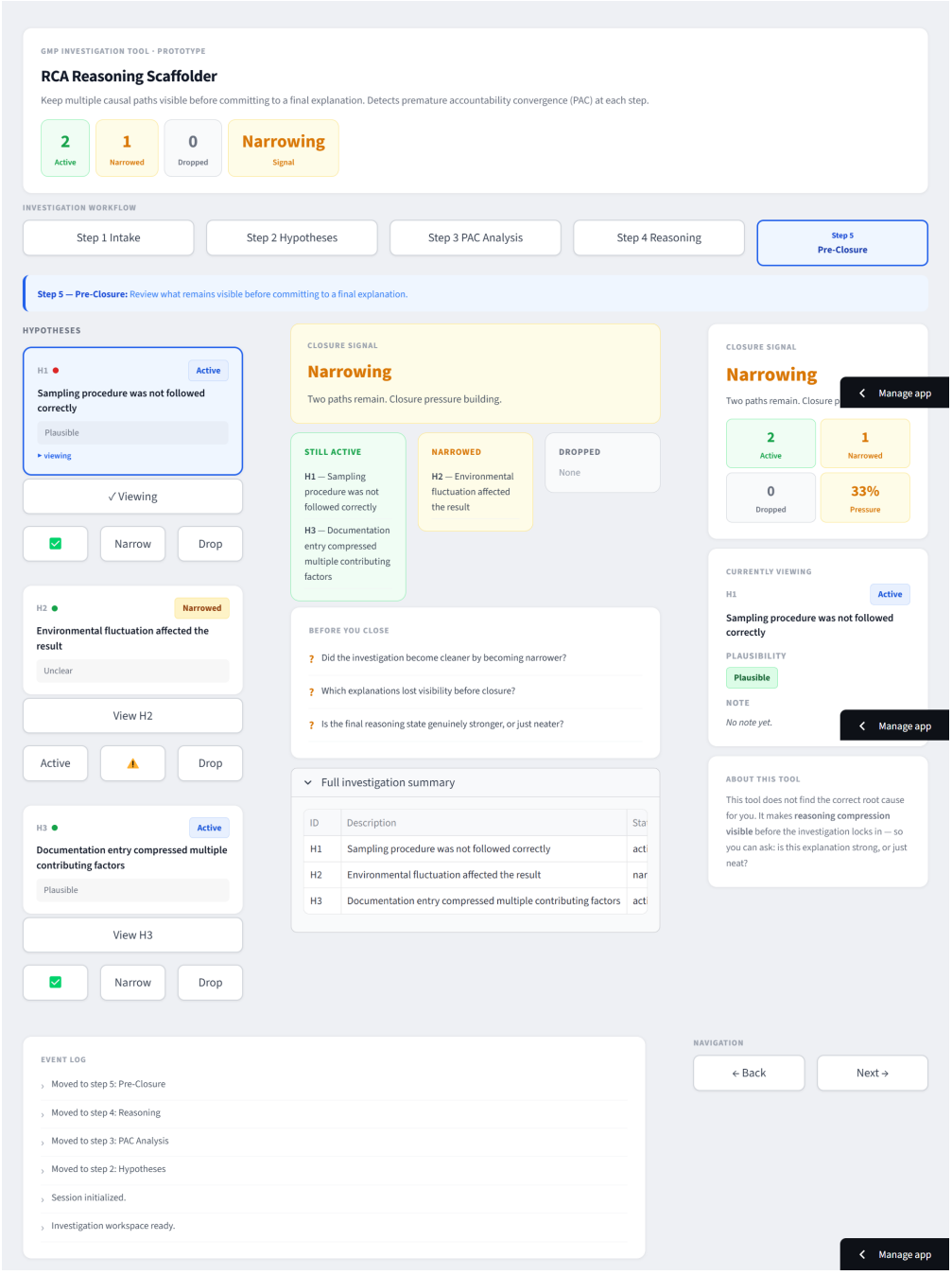



Figure 5: RCA Reasoning Scaffolder: Step 5 (Pre-Closure). Three verification questions are posed before closure is permitted: “Did the investigation become cleaner by becoming narrower?”, “Which explanations lost visibility before closure?”, and “Is the final reasoning state genuinely stronger, or just neater?”